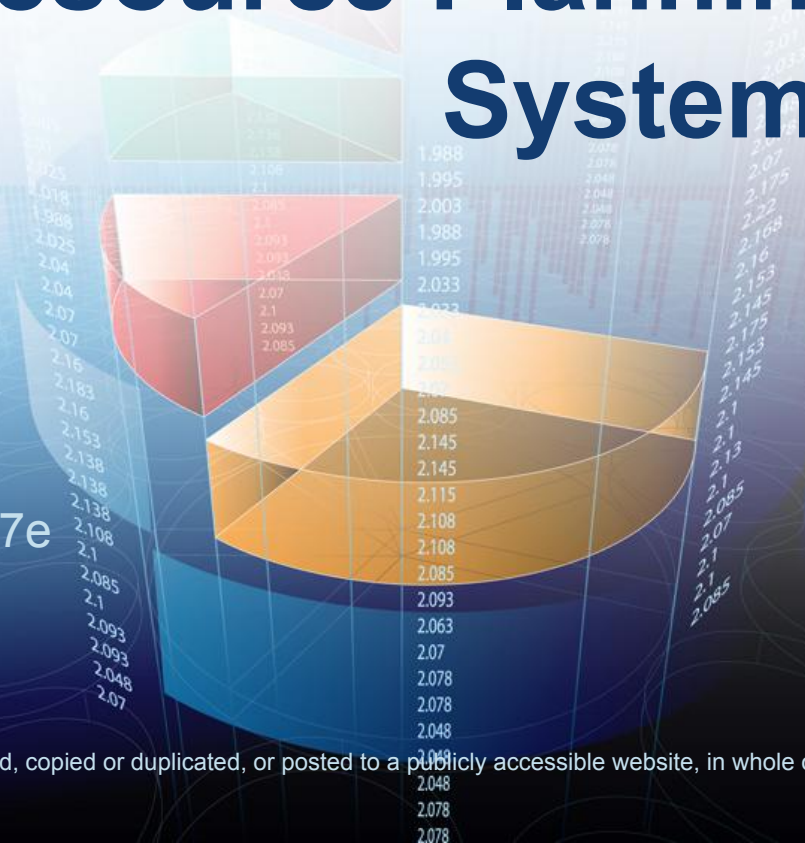
Chapter 11
Enterprise Resource Planning Systems



Accounting Information Systems, 7e
James A. Hall

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Objectives for Chapter 11

- Understand the general functionality and key elements of ERP systems.
- Understand the various aspects of ERP configurations including servers, databases, and the use of bolt-on software.
- Understand the purpose of data warehouses as a strategic tool and recognize the issues related to the design, maintenance, and operation of a data warehouse.
- Recognize the risks associated with ERP implementation.
- Be aware of key considerations related to ERP implementation.
- Understand the internal control and auditing implications associated with ERPs.

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Problems with Non-ERP Systems

- In-house design limits connectivity outside the company
- Tendency toward separate IS's within firm
  - lack of integration limits communication within the company
- Strategic decision-making not supported
- Long-term maintenance costs high
- Limits ability to engage in process reengineering

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Traditional IS Model: Closed Database Architecture

- Similar in concept to flat-file approach
  - data remains the property of the application
  - fragmentation limits communications
- Existence of numerous distinct and independent databases
  - redundancy and anomaly problems
- Paper-based
  - requires multiple entry of data
  - status of information unknown at key points

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Traditional Information System

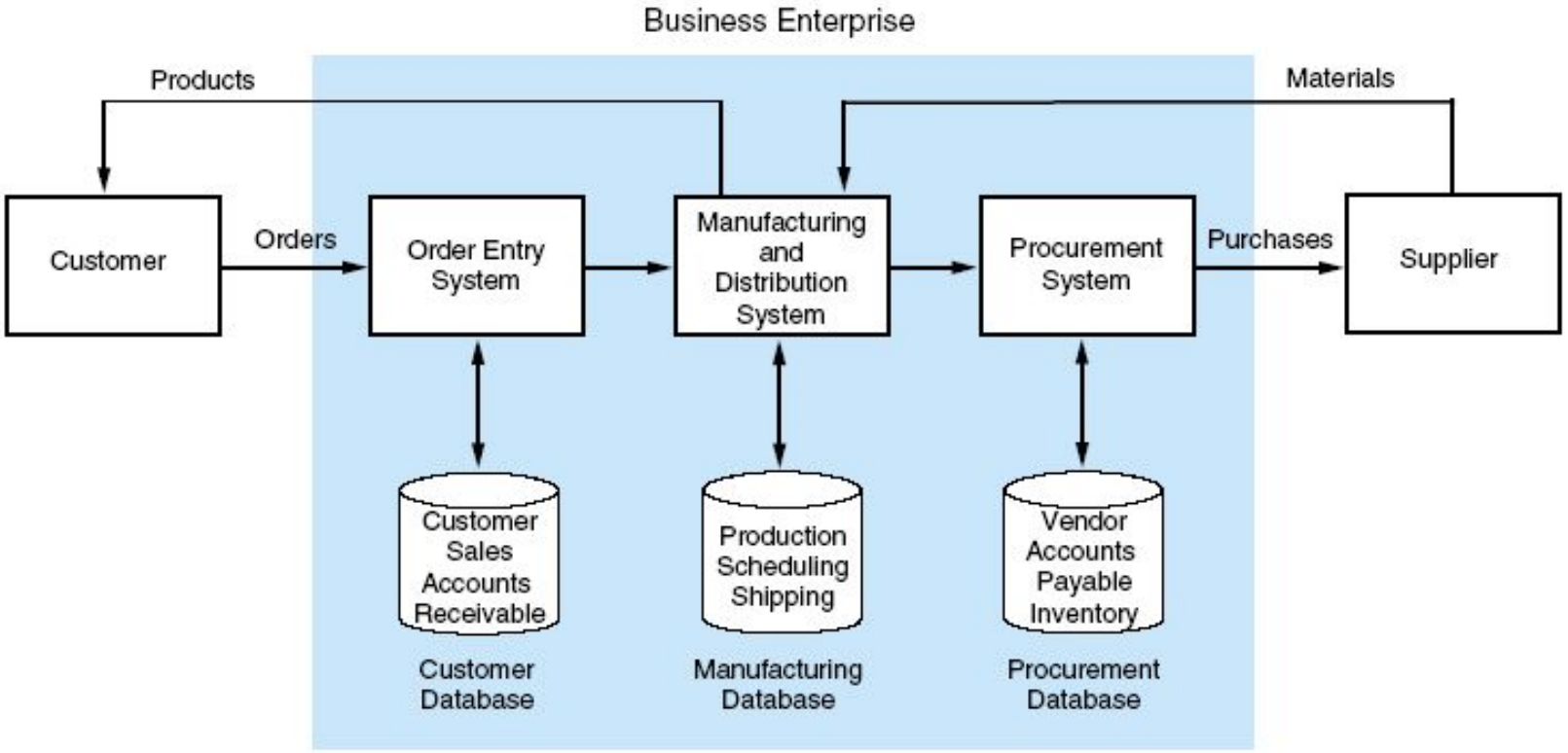


Figure 11-1

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What is an ERP System?

- Multi-module application software that helps a company manage the important parts of its business in an integrated fashion.
- Key features include:
  - smooth and seamless flow of information across organizational boundaries
  - standardized environment with shared database independent of applications and integrated applications

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ERP System

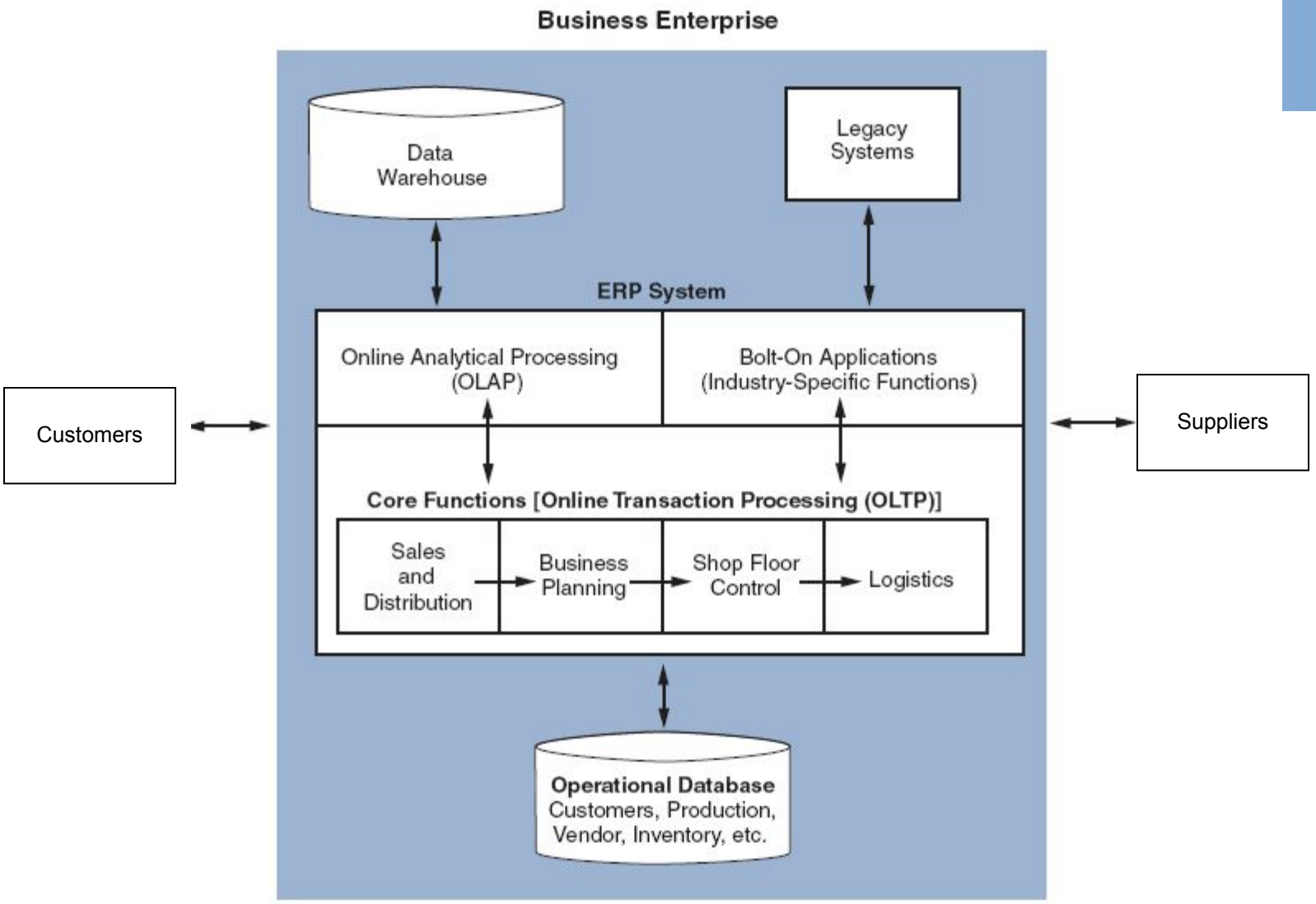


Figure 11-2

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Two Main ERP Applications

Core applications

- a.k.a. *On-line Transaction Processing (OLTP)*
- transaction processing systems
- support the day-to-day operational activities of the business
- support mission-critical tasks through simple queries of operational databases
- include sales and distribution, business planning, production planning, shop floor control, and logistics modules

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Two Main ERP Applications

Business analysis applications

- a.k.a. *On-line Analytical Processing (OLAP)*
- decision support tool for management-critical tasks through analytical investigation of complex data associations
- supplies management with “real-time” information and facilitates timely decisions to improve performance and achieve competitive advantage
- includes decision support, modeling, information retrieval, ad-hoc reporting/analysis, and what-if analysis

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OLAP

- Supports management-critical tasks through analytical investigation of complex data associations captured in data warehouses:
  - **Consolidation** is the aggregation or roll-up of data.
  - **Drill-down** allows the user to see data in selectively increasing levels of detail.
  - **Slicing and Dicing** enables the user to examine data from different viewpoints to uncover trends and patterns.

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ERP System Configurations: Client-Server Network Topology

Two-tier

- common server handles both application and database duties
- used especially in LANs

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Two-Tier Client Server

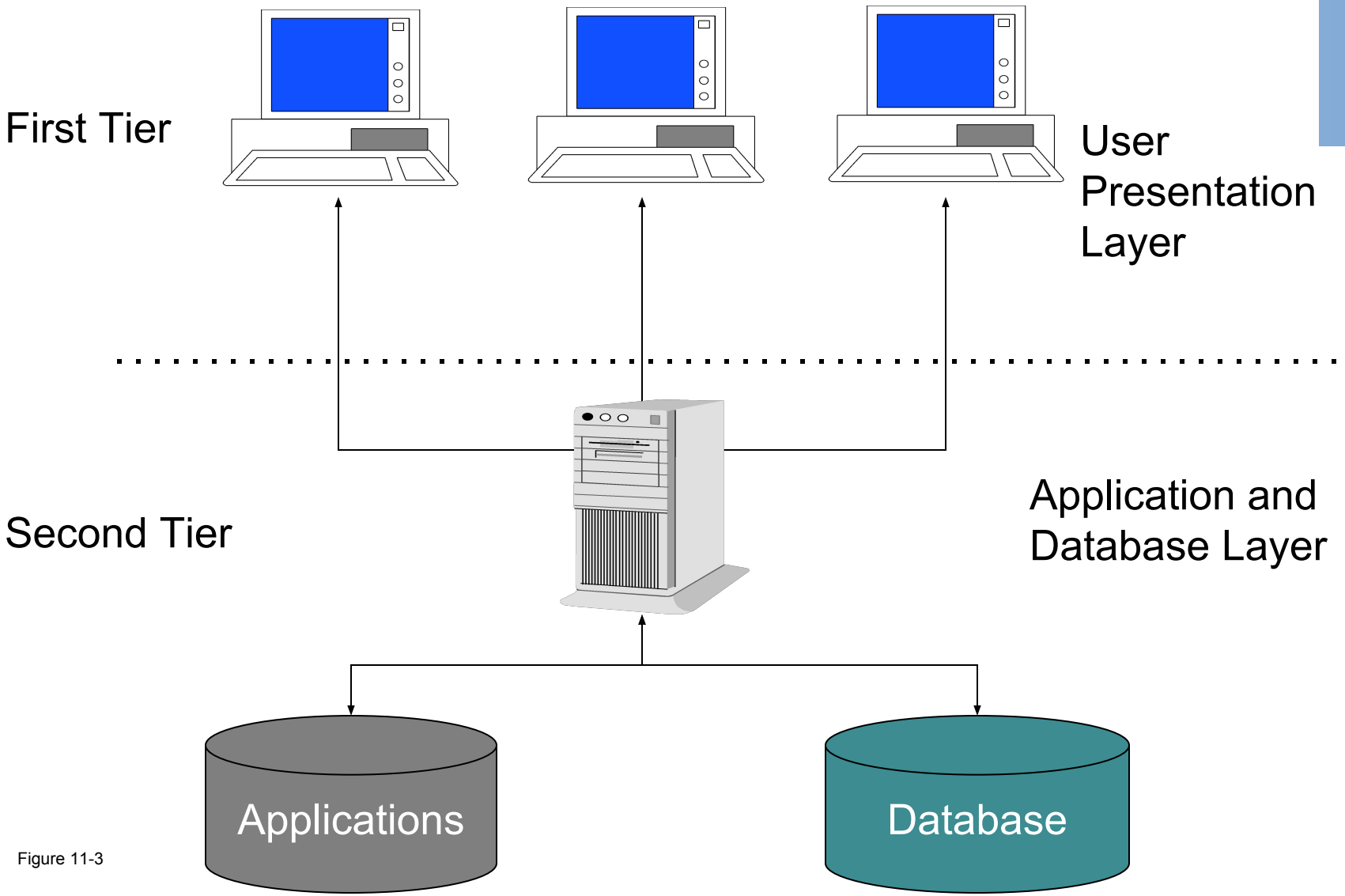


Figure 11-3

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ERP System Configurations: Client-Server Network Topology

Three-tier

- client links to the application server which then initiates a second connection to the database server
- used especially in WANs

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Three-Tier Client Server

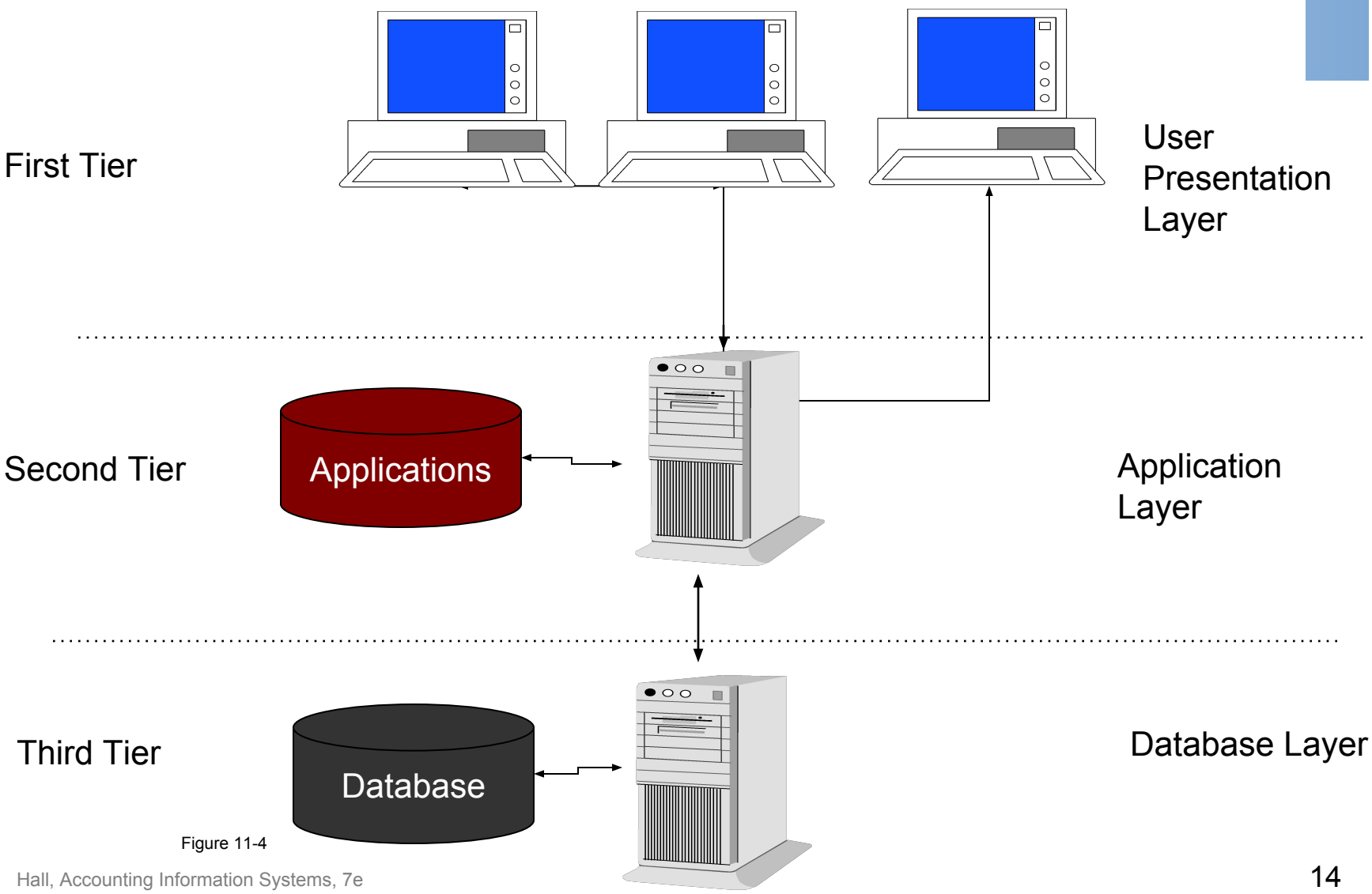


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OLTP and OLAP Client Server

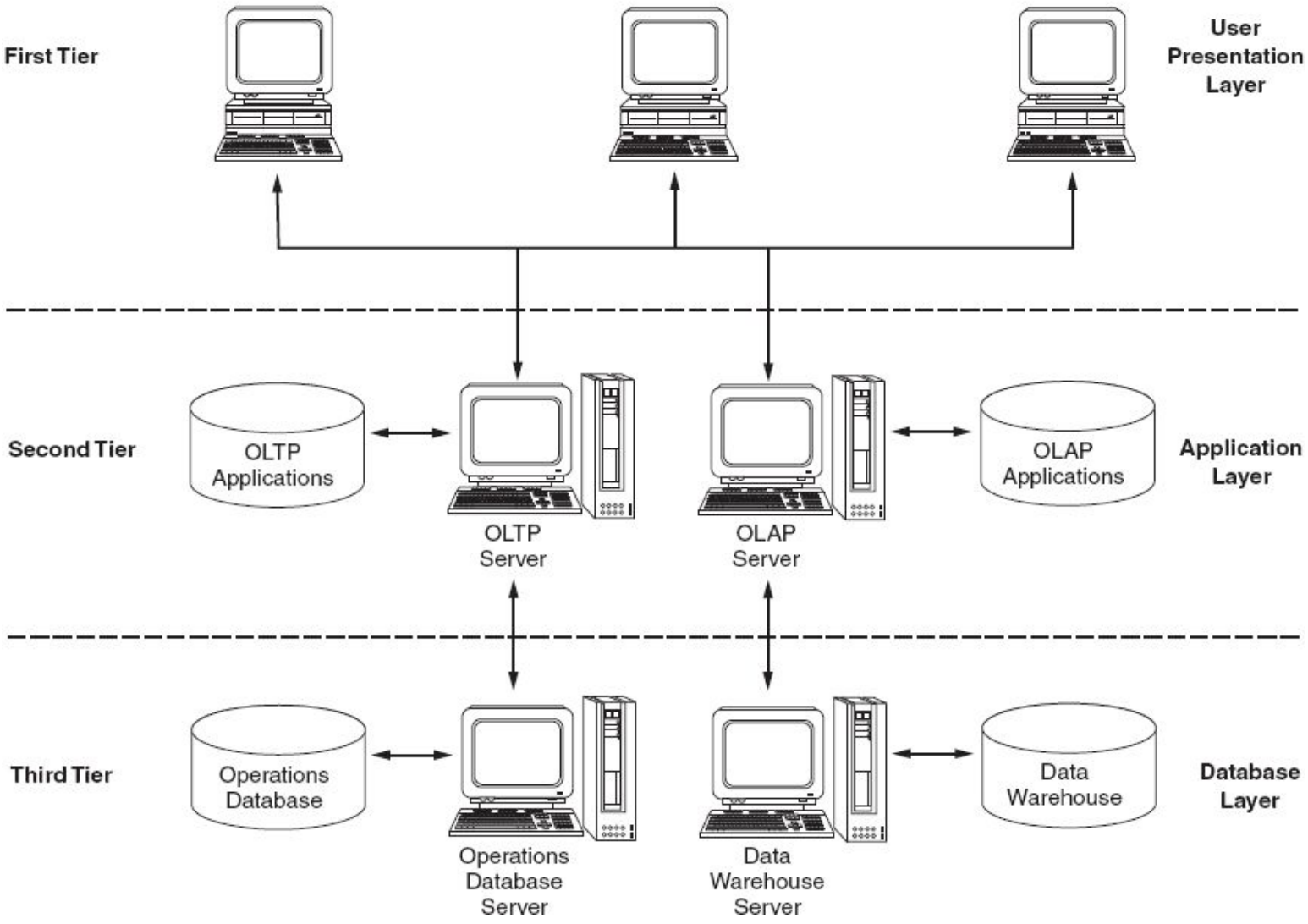


Figure 11-5

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ERP System Configurations: Databases and Bolt-Ons

- Database Configuration
  - selection of database tables in the thousands
  - setting the switches in the system
- Bolt-on Software
  - third-party vendors provide specialized functionality software
  - Supply Chain Management (SCM) links vendors, carriers, logistics companies, and IS providers

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What is a Data Warehouse?

- A *multi-dimensional* database often using hundreds of gigabytes or even terabytes of memory
  - Data are extracted periodically from operational databases or from public information services.
- A database constructed for quick searching, retrieval, ad-hoc queries, and ease of use
- ERP systems can exist without data warehouses.
  - However, most large ERP implementations include separate operational and data warehouse databases.
  - Otherwise, management data analysis may result in pulling system resources away from operational use.
  - Also, there are many sophisticated data-mining tools.

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Data Warehouse Process

The five stages of the data warehousing process:

1. modeling data for the data warehouse
2. extracting data from operational databases
3. cleansing extracted data
4. transforming data into the warehouse model
5. loading data into the data warehouse database

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Data Warehouse Process: Stage 1

- Modeling data for the data warehouse
  - Because of the vast size of a data warehouse, the warehouse database consists of de-normalized data.
    - Relational theory does not apply to a data warehousing system.
    - Normalized tables pertaining to selected events may be consolidated into de-normalized tables.

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Data Warehouse Process: Stage 2

- Extracting data from operational databases
  - The process of collecting data from operational databases, flat-files, archives, and external data sources.
  - Snapshots vs. stabilized data
    - A key feature of a data warehouse is that the data contained in it are in a non-volatile (stable) state.

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Data Warehouse Process: Stage 3

- Cleansing extracted data
  - Involves filtering out or repairing invalid data prior to being stored in the warehouse.
    - Operational data are “dirty” for many reasons: clerical, data entry, computer program errors, misspelled names and blank fields.
  - Also involves transforming data into standard business terms with standard data values.

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Data Warehouse Process: Stage 4

- Transforming data into the warehouse model
  - To improve efficiency, data are transformed into summary views before being loaded.
  - Unlike operational views, which are virtual in nature with underlying base tables, data warehouse views are physical tables.
    - OLAP permits users to construct virtual views.

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Data Warehouse Process: Stage 5

- Loading data into the data warehouse database
  - Data warehouses must be created & maintained separately from the operational databases.
    - internal efficiency
    - integration of legacy systems
    - consolidation of global data

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Data Warehouse System

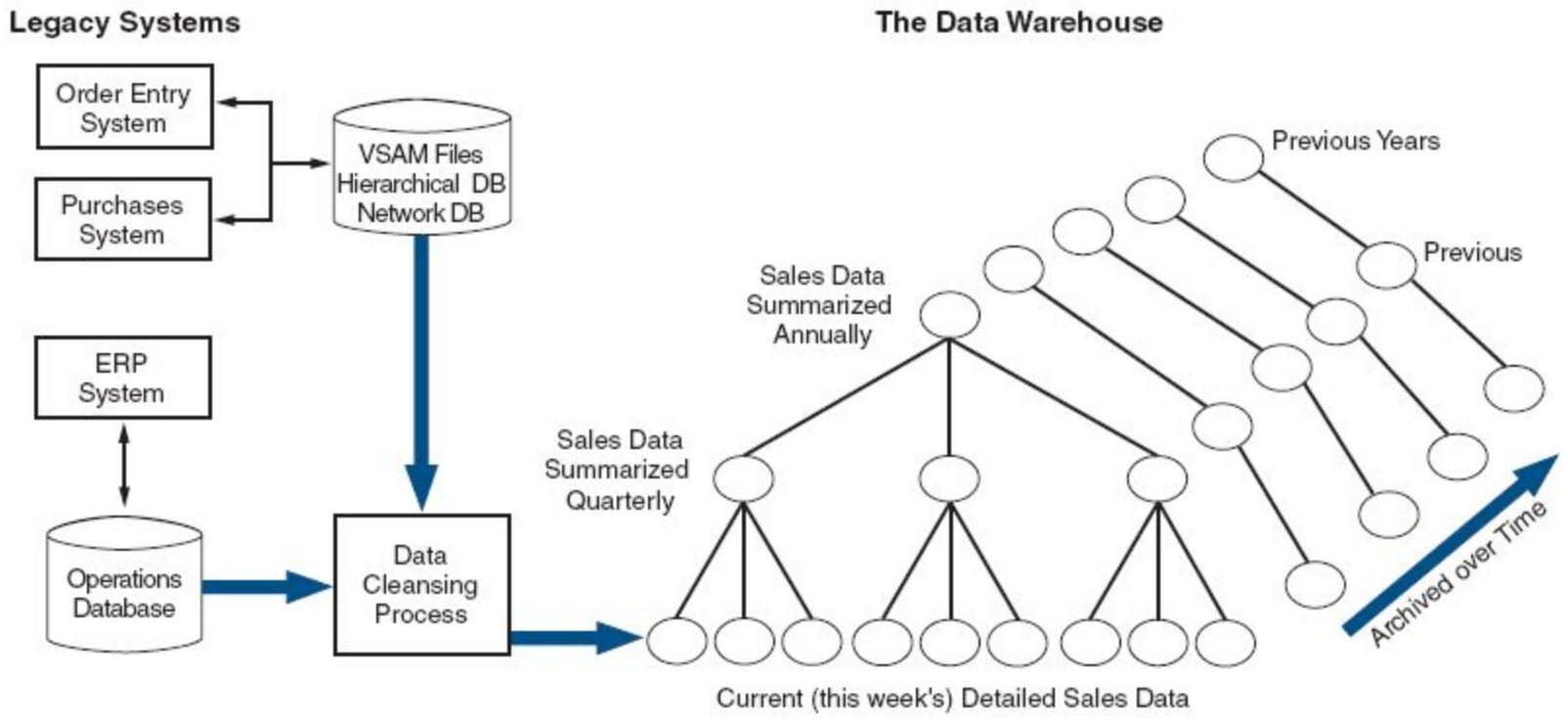


Figure 11-7

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Applications of Data Mining

Business Field	Application
Banking/Investments	Detect patterns of fraudulent credit card use.
	Identify loyal customers and predict those likely to change their credit card affiliation.
	Examine historical market data to determine investors' stock trading rules.
	Predict credit card spending of key customer groups.
	Identify correlations between different financial indicators.
Health Care and Medical Insurance	Predict office visits from historical analysis of historical patient behavior.
	Identify successful and economic medical therapies for different illnesses.
	Identify which medical procedures tend to be claimed together.
	Predict which customers will buy new policies.
	Identify behavior patterns associated with high-risk customers.
	Identify indicators of fraudulent behavior.
Marketing	Identify buying patterns based on historical customer data.
	Identify relationships among customer demographic data.
	Predict response to various forms of marketing and promotion campaigns.

Table 11-1

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Risks Associated with ERP Implementation

- Pace of implementation
  - 'Big Bang'--switch operations from legacy systems to ERP in a single event
  - 'Phased-In'--independent ERP units installed over time, assimilated, and integrated
- Opposition to change
  - user reluctance and inertia
  - need of upper management support

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Risks Associated with ERP Implementation

- Choosing the wrong ERP
  - goodness of fit: no one ERP product is best for all industries
  - scalability: system's ability to grow
- Choosing the wrong consultant
  - common to use a third-party (the Big Four)
  - thoroughly interview potential consultants
  - establish explicit expectations

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Risks Associated with ERP Implementation

- High cost and cost overruns
  - common areas with high costs:
    - training
    - testing and integration
    - database conversion
- Disruptions to operations
  - ERP implementations usually involve business process reengineering (BPR)
    - expect major changes in business processes

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Implications for Internal Control and Auditing

- Transaction authorization
  - Controls are needed to validate transactions before they are accepted by other modules.
  - ERPs are more dependent on programmed controls than on human intervention.
- Segregation of duties
  - Manual processes that normally require segregation of duties are often eliminated.
  - *User role:* predefined user roles limit a user's access to certain functions and data.

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Implications for Internal Control and Auditing

- Supervision
  - Supervisors need to acquire a technical and operational understanding of the new system.
  - Employee-empowered philosophy should not eliminate supervision.
- Accounting records
  - Corrupted data may be passed from external sources and from legacy systems.
  - loss of paper audit trail

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Implications for Internal Control and Auditing

- Access controls
  - critical concern with confidentiality of information
  - Who should have access to what?
- Access to data warehouse
  - Data warehouses often involve sharing information with suppliers and customers.

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Implications for Internal Control and Auditing

- Contingency planning
  - keeping a business going in case of disaster
  - key role of servers requires backup plans: redundant servers or shared servers
- Independent verification
  - traditional verifications are meaningless
  - need to shift from transaction level to overall performance level

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